PROJECT REPORT

ISTR

Intergalactic Space Travel Request System

A ServiceNow Enterprise Workflow Automation Application

Submitted in partial fulfilment of the requirements for
Bachelor of Technology
in Computer Science Engineering (AI & ML)

Submitted by
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1. Introduction

1.1 Project Overview

The Intergalactic Space Travel Request (ISTR) System is a ServiceNow-based enterprise application designed to automate the complete lifecycle of intergalactic mission requests. It replaces manual paperwork with a centralized digital workflow where mission requesters can submit travel requests, upload required documents, obtain multi-level approvals, receive automated notifications, monitor mission status, and generate reports through dashboards.

The application is built on ServiceNow App Engine Studio as a scoped application, using a custom table (x_2065295_istr_i_space_travel_request) that extends the base Task table, and leverages native ServiceNow capabilities — Flow Designer, Business Rules, Client Scripts, UI Policies, ACLs, Notifications, Reports, and Performance Analytics — to deliver a secure, transparent, and auditable mission management system.

1.2 Purpose

The purpose of ISTR is to improve the efficiency, transparency, security, and traceability of space mission request management by eliminating manual, email-based, and paper-based coordination between Mission Requesters, Managers, Finance Officers, and Mission Control. The system aims to:

- Automate mission request submission and approval routing
- Eliminate manual paperwork and repetitive follow-ups
- Enforce role-based security across all mission data
- Provide real-time visibility through notifications, reports, and dashboards

2. Ideation Phase

2.1 Problem Statement

Traditional mission request processes rely heavily on manual documentation, email communication, and multiple disconnected approval stages, resulting in a slow approval process, lack of transparency, poor tracking, repetitive paperwork, communication delays, difficult reporting, and security concerns.

Customer Problem Statements

Persona	Need	Problem
Mission Requester	Faster mission submission and tracking	Manual approvals with no visibility into status
Mission Control	Centralized approval management	Scattered information across departments and emails

2.2 Empathy Map Canvas

An empathy map was built for the primary persona — the Mission Requester — to understand their experience of the existing manual process.

Dimension	Insight
Think	Will my request be approved?
Hear	“Please wait for approval.”
See	Multiple, disconnected approval levels
Say	“I need to track my request.”
Pain	Manual process, no visibility, repeated follow-ups
Gain	An automated, transparent workflow

2.3 Brainstorming

A structured brainstorming session was conducted around the question: “How might we simplify and accelerate the intergalactic mission request and approval process while maintaining transparency and control?” Ideas were grouped into themes and prioritized by impact and effort.

Idea Groups

- Automated multi-level approval routing (Manager → Finance → Mission Control)
- Centralized mission request catalog with structured intake form
- Real-time status tracking and automated email notifications
- Role-based dashboards for requesters, approvers, and administrators
- Conditional fields for special mission types (e.g. Alien Negotiation → Diplomatic Officer)

Idea Prioritization Matrix

Idea	Impact	Effort	Priority
Multi-level approval workflow (Flow Designer)	High	Medium	P1
Mission request catalog item	High	Low	P1

Idea	Impact	Effort	Priority
ACL-based role security (10 ACLs)	High	Medium	P1
Email notifications	Medium	Low	P1
Reports & Mission Analytics dashboard	Medium	Medium	P2
Conditional UI Policy / Client Script validation	Medium	Low	P2

3. Requirement Analysis

3.1 Customer Journey Map

The journey map traces a requester's experience from mission conception through mission history, across the ISTR system.

Stage	Action
Awareness	Requester identifies the need for a mission
Service Catalog	Requester locates the Mission Request item
Mission Request Form	Requester fills in mission details
Submit Request	Form submitted; record created
Manager Approval	First-level review and approval
Finance Approval	Budget review and approval
Mission Control	Final approval and scheduling
Mission Scheduled	Mission confirmed and resourced
Mission Completed	Mission executed and closed
History	Record retained for reporting and audit

3.2 Solution Requirement

Functional Requirements

- Mission Request – submission, editing, cancellation, and viewing
- Approval Workflow – Manager, Finance, and Mission Control approval stages
- Workflow Automation – implemented using Flow Designer
- User Management – Roles, Groups, and ACL-based access
- Mission Tracking – real-time status across the mission lifecycle
- Reports – Mission Status, Approval Summary, Risk Analysis, Monthly Requests
- Dashboard – Mission Analytics via Performance Analytics
- Business Rules, Client Scripts, and UI Policies – validation and dynamic forms

Non-Functional Requirements

- Usability – guided, single-page mission request form
- Security – role-based access enforced via 10 ACLs
- Performance – fast form load and flow execution on the PDI
- Availability – hosted on ServiceNow SaaS cloud infrastructure
- Reliability – consistent approval routing and notification delivery
- Scalability – scoped application design supports future module growth
- Maintainability – configuration-based logic (Business Rules, Flow Designer) over hard-coded scripts

3.3 Data Flow Diagram

Data flows from the Mission Requester through the Mission Form, Business Rules, the custom table, Flow Designer approval stages, and finally into Notifications and the Dashboard.

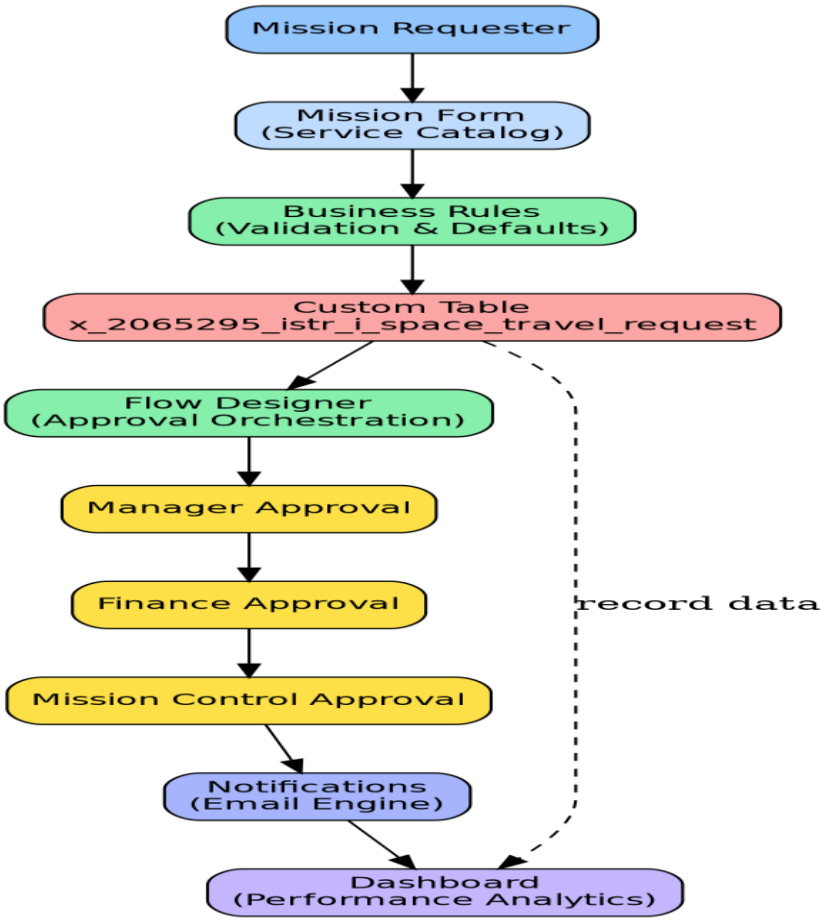


Figure 3.1: ISTR Data Flow Diagram

3.4 Technology Stack

Layer	**Technology**
Frontend	Service Portal, Service Catalog
Backend	Business Rules, Flow Designer, Client Scripts, UI Policies
Database	ServiceNow Custom Table (x_2065295_istr_i_space_travel_request, extends Task)
Security	ACLs, Users, Roles, Groups
Notifications	Email Engine
Reports	ServiceNow Reports
Dashboard	Performance Analytics
Cloud	ServiceNow SaaS

4. Project Design

4.1 Problem Solution Fit

Element	**Details**
Customer	Scientists, Astronauts, Mission Managers
Jobs	Mission Requests, Approval, Tracking
Available Solutions	Paper Forms, Emails, Excel
Customer Constraints	Manual approvals, department dependency, poor visibility
Behaviour	Manual submissions, email follow-ups
Triggers	Mission planning, launch schedules
Solution	ISTR ServiceNow Application

4.2 Proposed Solution

ISTR is a centralized ServiceNow application that provides:

- Online Mission Requests – submitted through the Service Catalog
- Automated Approval Workflow – orchestrated via Flow Designer
- Role-Based Access – enforced through ACLs, Roles, and Groups
- Email Notifications – submitted, approved, rejected, completed
- Dashboard – Mission Analytics via Performance Analytics
- Reports – Mission Status, Approval Summary, Risk Analysis, Monthly Requests
- Mission Tracking – full lifecycle visibility from Draft to Closed
- Secure Data Management – custom table extending Task, protected by 10 ACLs

4.3 Solution Architecture

ISTR follows a layered architecture spanning frontend, backend/logic, security, database, notification, and analytics tiers, all on the ServiceNow SaaS platform.

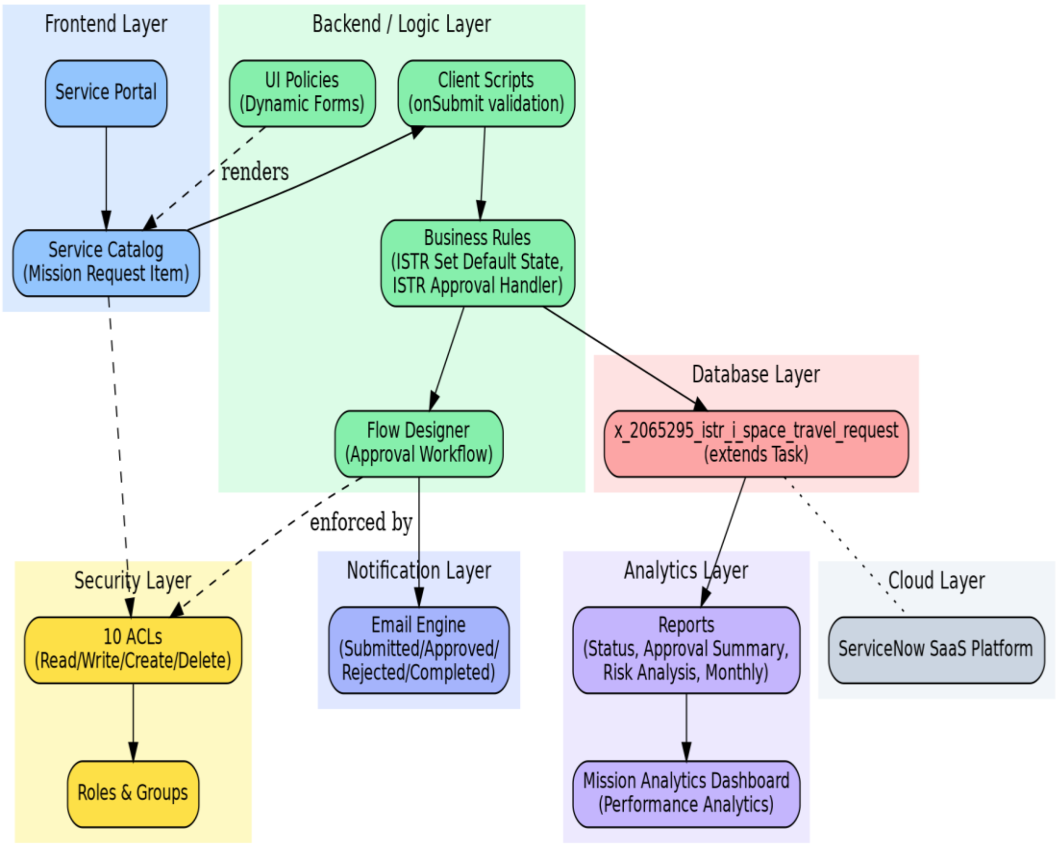


Figure 4.1: ISTR Solution Architecture

Database Design

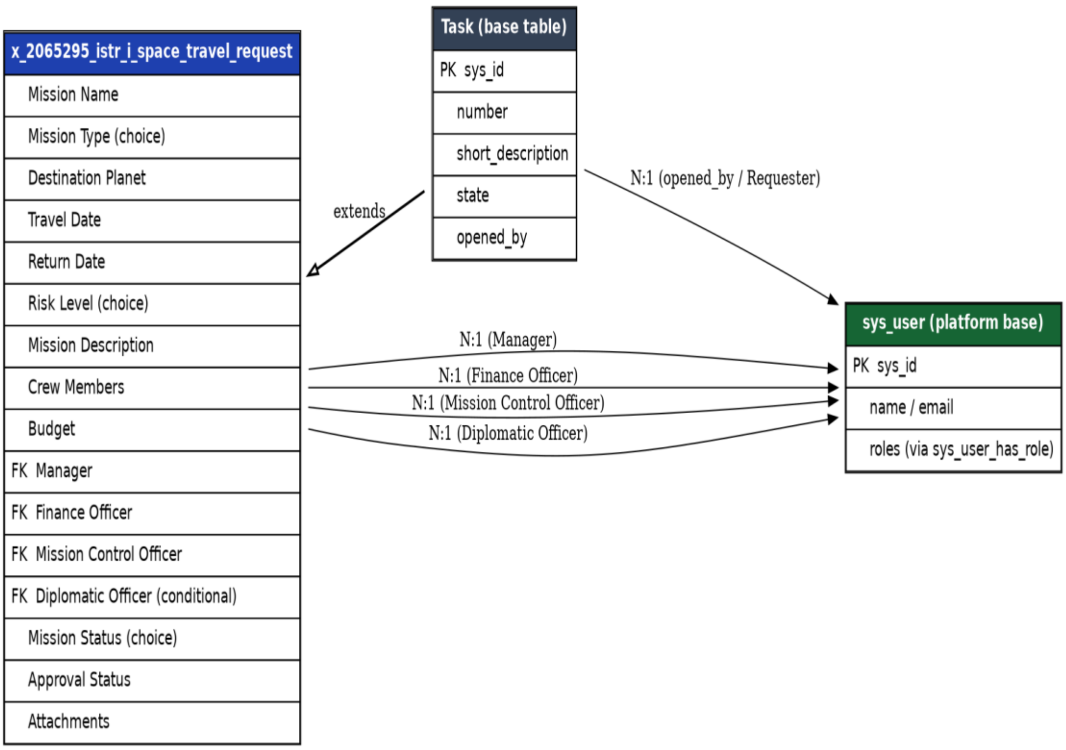


Figure 4.2: ISTR Entity-Relationship Diagram

Field	Type / Notes
Mission ID	Auto-numbered (inherited from Task)
Mission Name	String
Mission Type	Choice: Research, Exploration, Colonization, Emergency Rescue, Cargo Transport, Diplomatic Mission, Alien Negotiation, Military Operation
Destination Planet	String / reference
Travel Date / Return Date	Date/Time
Risk Level	Choice: Low, Medium, High, Critical
Mission Description	Multi-line text
Crew Members	List / reference
Budget	Currency
Manager / Finance Officer / Mission Control Officer / Diplomatic Officer	Reference (sys_user)
Mission Status	Choice: Draft, Submitted, Manager Approved, Finance Approved, Mission Control Approved, Rejected, Scheduled, Completed, Cancelled
Approval Status	Choice / rollup
Attachments	Standard attachment relationship

Security Model

Roles	Groups
Mission Requester	Mission Requesters
Manager	Managers
Finance Officer	Finance Team
Mission Control	Mission Control Team
System Administrator	System Administrators

10 ACLs secure Create, Read, Write, and Delete operations on the ISTR table across all five roles.

Automation Layer

Component	Details
Business Rule: ISTR Set Default State	Automatically sets the initial request state on creation
Business Rule: ISTR Approval Handler	Handles approval status updates as the request moves through workflow stages
Client Script (onSubmit)	Validates that Alien Negotiation missions specify a Diplomatic Officer
UI Policy	Shows Diplomatic Officer field only when Mission Type = Alien Negotiation

Approval Workflow (Flow Designer)

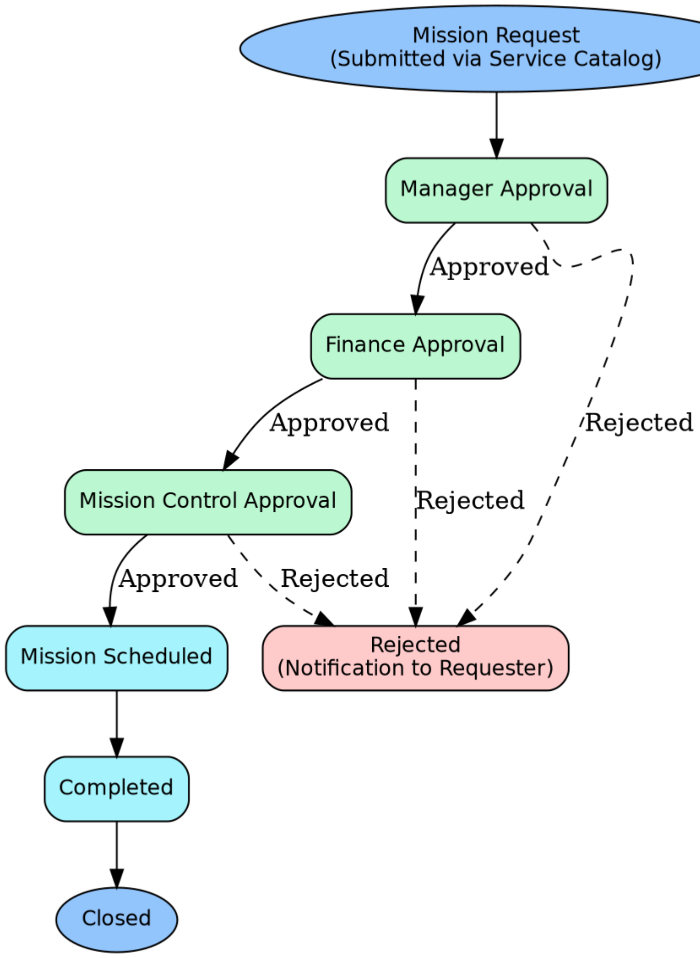


Figure 4.3: ISTR Approval Workflow

Notifications

- Submission – confirmation email to requester
- Approval – notification at each approval stage
- Rejection – notification with reason
- Completion – notification to requester and Mission Control

Analytics Layer

Dashboard widgets: Pending Requests, Approved Requests, Rejected Requests, Mission Types, Risk Levels, Monthly Requests.

Reports: Mission Status, Approval Summary, Risk Analysis, Monthly Requests.

5. Project Planning & Scheduling

5.1 Project Planning

Planning followed Agile/Scrum concepts: a Sprint is a fixed period (6 days) for completing a set of tasks; an Epic is a large body of work broken into Stories; a Story Point (Fibonacci-style: 1, 2, 3, 5) estimates the relative effort of a Story.

Product Backlog

Sprint	Epic	US No.	User Story	SP	Priority
Sprint-1	Mission Request Management	USN-1	Submit a mission request with destination, dates, crew, budget	3	High
Sprint-1		USN-2	Save a request as Draft before submission	2	Medium
Sprint-1		USN-3	Attach supporting mission documents	2	Medium
Sprint-1	Approval Workflow	USN-4	Manager reviews and approves/rejects a request	3	High
Sprint-1		USN-5	Finance Officer approves budget for flagged missions	3	High
Sprint-2	Approval Workflow	USN-6	Mission Control gives final approval and schedules mission	3	High
Sprint-2	Security	USN-7	Admin configures Roles and Groups	2	Medium
Sprint-2		USN-8	Admin creates ACLs to secure record access by role	3	High
Sprint-2	Dynamic Forms	USN-9	Diplomatic Officer field shown only for Alien Negotiation missions	2	Medium
Sprint-3	Notifications	USN-10	Requester receives emails on all state changes	3	High
Sprint-3	Dashboard	USN-11	Manager views Mission Analytics dashboard	5	High
Sprint-3	Reports	USN-12	Admin generates Mission Status, Approval, and Risk reports	3	Medium

Sprint-wise Story Points & Velocity

- Sprint 1 Total = 3+2+2+3+3 = 13
- Sprint 2 Total = 3+2+3+2 = 10
- Sprint 3 Total = 3+5+3 = 11

Total Story Points = 34 | Number of Sprints = 3 | Velocity = 34 ÷ 3 ≈ 11.3 SP/Sprint

Project Tracker

Sprint	Total SP	Duration	Start Date	End Date (Planned)	SP Completed	Release Date (Actual)
Sprint-1	13	6 Days	01 Jun 2026	06 Jun 2026	13	06 Jun 2026
Sprint-2	10	6 Days	08 Jun 2026	13 Jun 2026	10	13 Jun 2026
Sprint-3	11	6 Days	15 Jun 2026	20 Jun 2026	11	20 Jun 2026

Burndown Chart

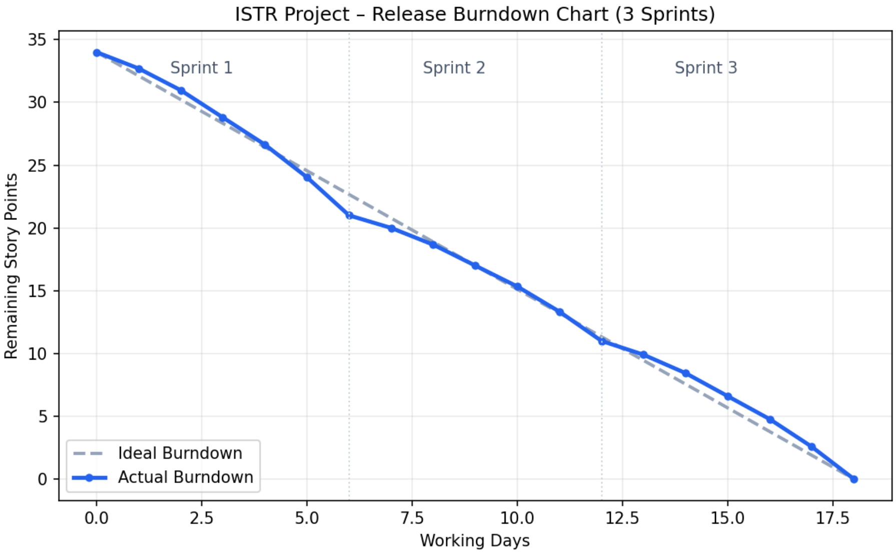


Figure 5.1: ISTR Release Burndown Chart

6. Functional and Performance Testing

6.1 Performance Testing

Functional and performance testing was carried out on the ISTR Service Catalog form, approval workflow, and record-level security, on the Personal Developer Instance.

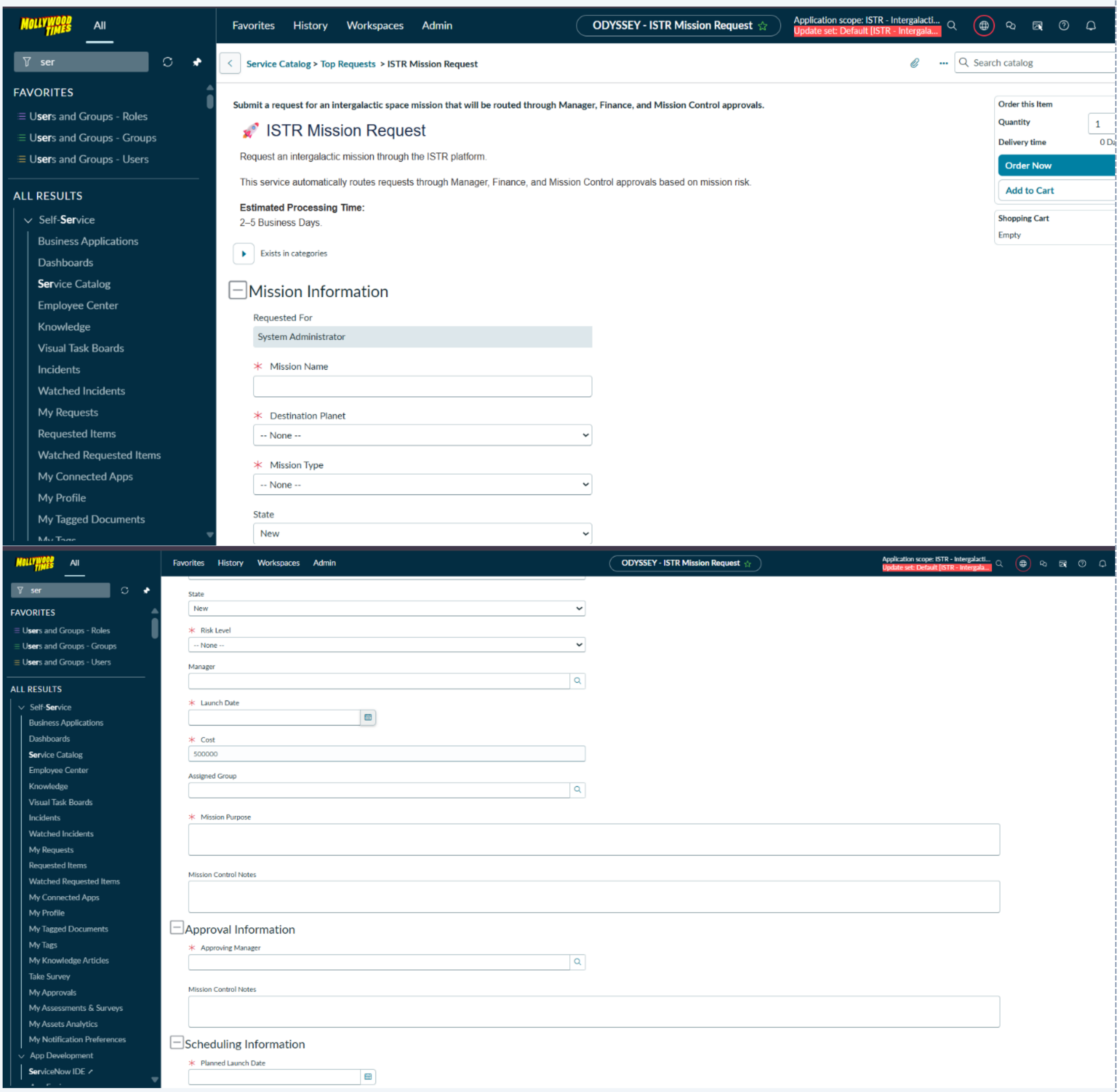
Test ID	Scenario	Expected Result	Actual Result	Pass/Fail
FT-01	Text input validation (Mission Name, Destination)	Valid inputs accepted; error for blanks	Error shown correctly	Pass
FT-02	Crew size vs spacecraft/date validation	Blocked for invalid values	Blocked correctly	Pass
FT-03	Mission request submission	Record created, state = Submitted	Created as expected	Pass
FT-04	Multi-level approval workflow	Sequential routing Manager→Finance→Mission Control	Executed in correct sequence	Pass
FT-05	ACL / role-based access	Access denied to other users' records	Denied correctly	Pass
FT-06	Notification delivery	Email at each state change	Emails received at each stage	Pass
PT-01	Catalog form load time	Under 2 seconds	~1.4 seconds	Pass
PT-02	Approval flow execution time	Triggers within seconds	~2 seconds	Pass
PT-03	Attachment upload load test	No errors/slowdown	Uploaded successfully	Pass

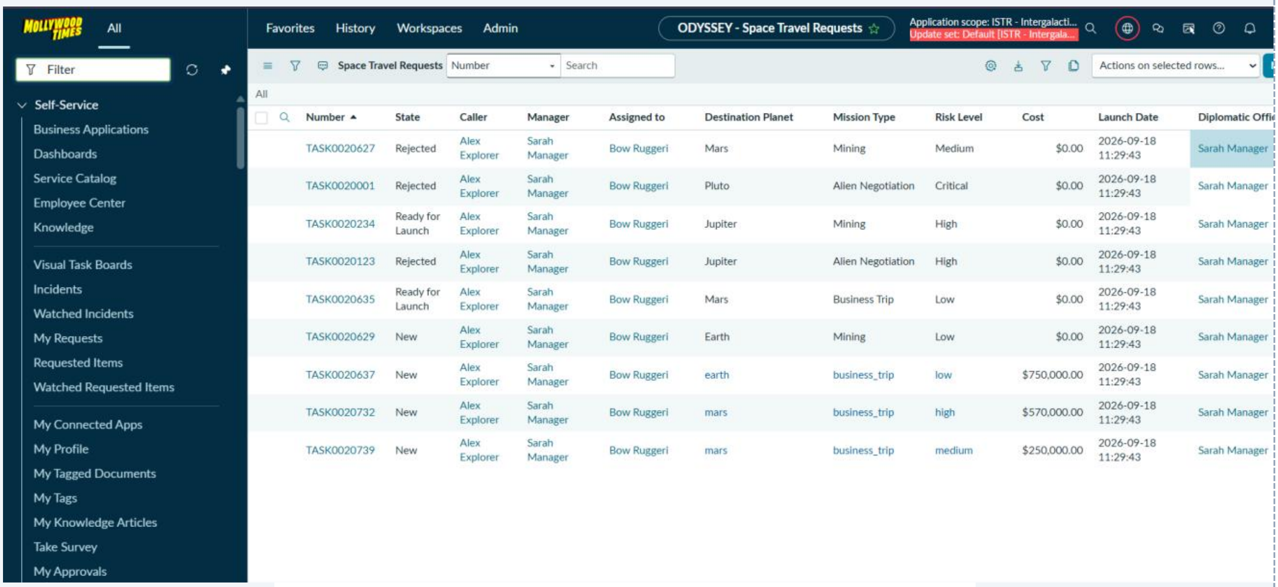
User Acceptance Testing (UAT) was additionally conducted with 6 test cases (TC-001 to TC-006) covering submission, approval, rejection, role-based visibility, and dashboard accuracy — all passed, with 2 minor bugs found and resolved (UI Policy field visibility, duplicate notification trigger).

7. Results

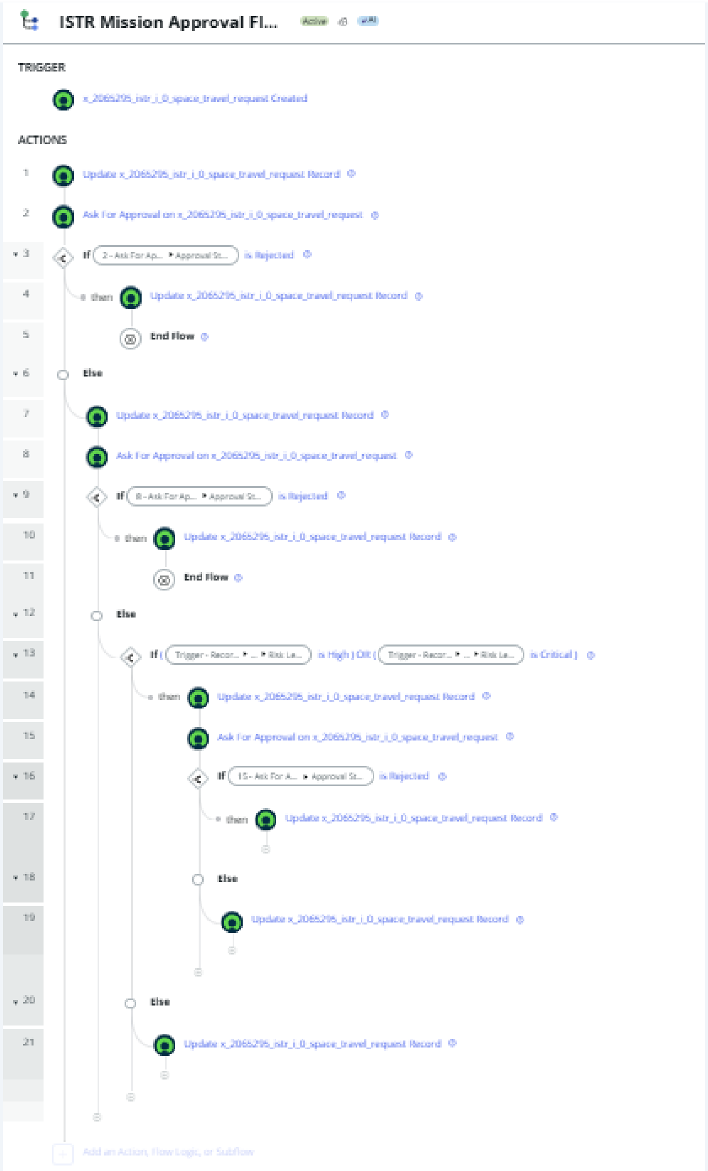
7.1 Output Screenshots

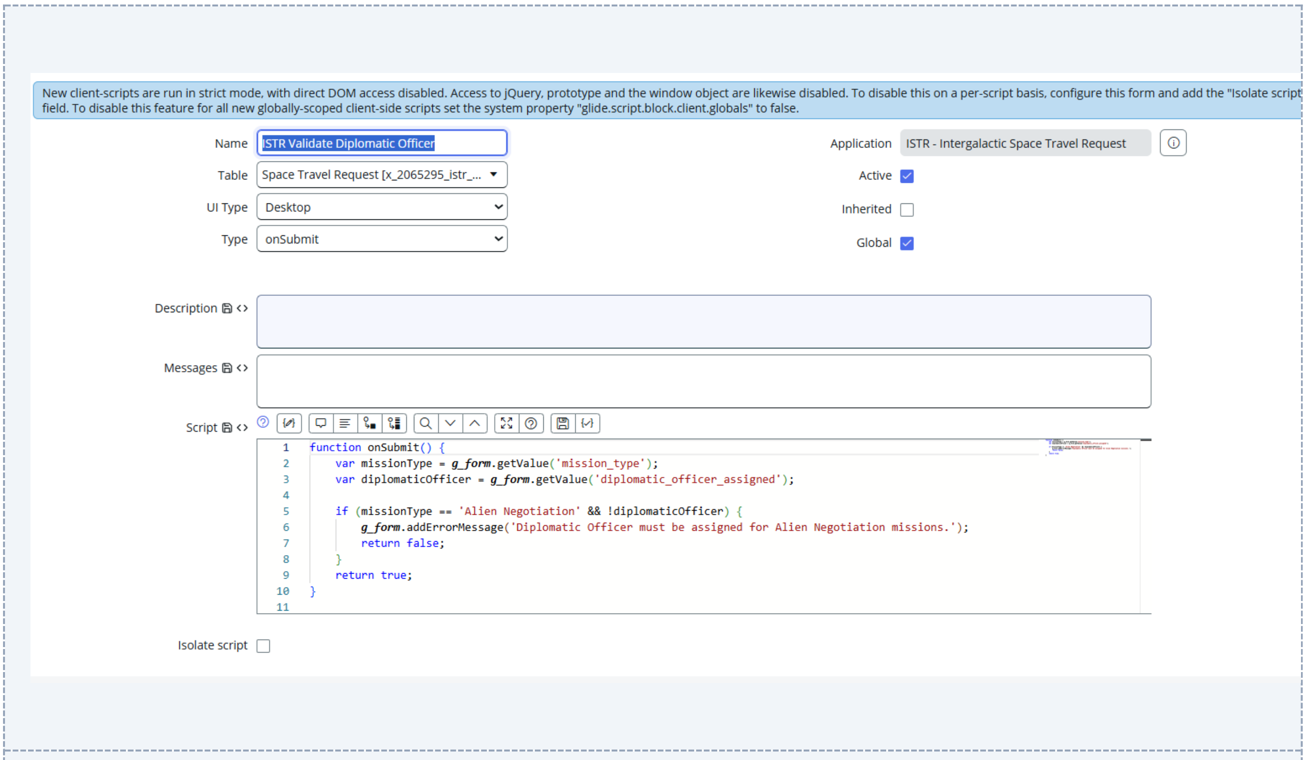
The following screenshots should be captured from the ServiceNow Personal Developer Instance and inserted here to evidence a working system. See the checklist provided separately for exact navigation steps.





Number	State	Caller	Manager	Assigned to	Destination Planet	Mission Type	Risk Level	Cost	Launch Date	Diplomatic Offi
TASK0020627	Rejected	Alex Explorer	Sarah Manager	Bow Ruggeri	Mars	Mining	Medium	$0.00	2026-09-18 11:29:43	Sarah Manager
TASK0020001	Rejected	Alex Explorer	Sarah Manager	Bow Ruggeri	Pluto	Alien Negotiation	Critical	$0.00	2026-09-18 11:29:43	Sarah Manager
TASK0020234	Ready for Launch	Alex Explorer	Sarah Manager	Bow Ruggeri	Jupiter	Mining	High	$0.00	2026-09-18 11:29:43	Sarah Manager
TASK0020123	Rejected	Alex Explorer	Sarah Manager	Bow Ruggeri	Jupiter	Alien Negotiation	High	$0.00	2026-09-18 11:29:43	Sarah Manager
TASK0020635	Ready for Launch	Alex Explorer	Sarah Manager	Bow Ruggeri	Mars	Business Trip	Low	$0.00	2026-09-18 11:29:43	Sarah Manager
TASK0020629	New	Alex Explorer	Sarah Manager	Bow Ruggeri	Earth	Mining	Low	$0.00	2026-09-18 11:29:43	Sarah Manager
TASK0020637	New	Alex Explorer	Sarah Manager	Bow Ruggeri	earth	business_trip	low	$750,000.00	2026-09-18 11:29:43	Sarah Manager
TASK0020732	New	Alex Explorer	Sarah Manager	Bow Ruggeri	mars	business_trip	high	$570,000.00	2026-09-18 11:29:43	Sarah Manager
TASK0020739	New	Alex Explorer	Sarah Manager	Bow Ruggeri	mars	business_trip	medium	$250,000.00	2026-09-18 11:29:43	Sarah Manager




New client-scripts are run in strict mode, with direct DOM access disabled. Access to jQuery, prototype and the window object are likewise disabled. To disable this on a per-script basis, configure this form and add the "Isolate script" field. To disable this feature for all new globally-scoped client-side scripts set the system property "glide.script.block.client.globals" to false.
Name
ISTR Validate Diplomatic Officer
Table
Space Travel Request [x_2065295_istr_...
UI Type
Desktop
Type
onSubmit
Application
ISTR - Intergalactic Space Travel Request
Active
Inherited
Global
Description
Messages
Script
function onSubmit() {
var missionType = g_form.getValue('mission_type');
var diplomaticOfficer = g_form.getValue('diplomatic_officer_assigned');
if (missionType == 'Alien Negotiation' && !diplomaticOfficer) {
g_form.addErrorMessage('Diplomatic Officer must be assigned for Alien Negotiation missions.');
return false;
}
return true;
}
Isolate script

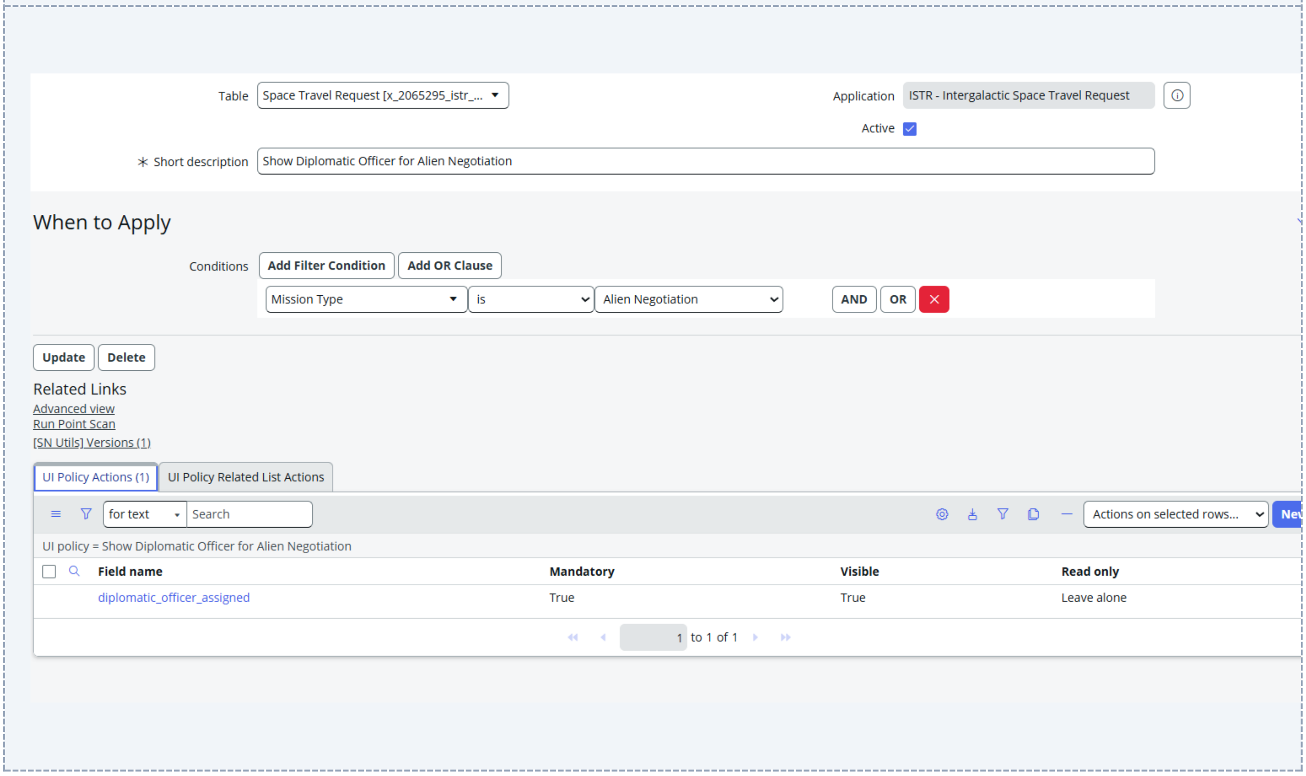
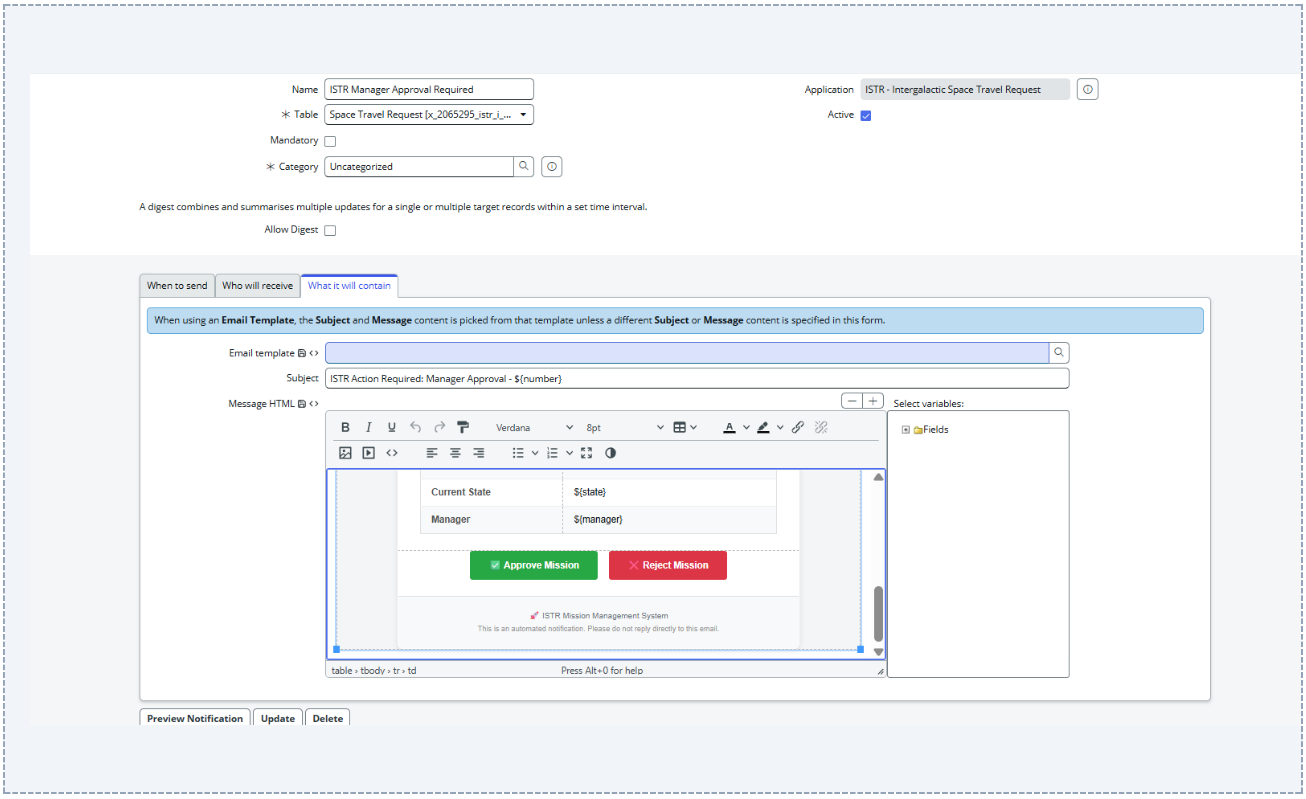


Table
Space Travel Request [x_2065295_istr_...
Application
ISTR - Intergalactic Space Travel Request
Active
Short description
Show Diplomatic Officer for Alien Negotiation
When to Apply
Conditions
Add Filter Condition
Add OR Clause
Mission Type
is
Alien Negotiation
AND
OR
Update
Delete
Related Links
Advanced view
Run Point Scan
[SN Utils] Versions (1)
UI Policy Actions (1)
UI Policy Related List Actions
for text
Search
Actions on selected rows...
UI policy = Show Diplomatic Officer for Alien Negotiation
Field name
Mandatory
Visible
Read only
diplomatic_officer_assigned
True
True
Leave alone
1 to 1 of 1


Name
ISTR Manager Approval Required
Table
Space Travel Request [x_2065295_istr_i_...
Mandatory
Category
Uncategorized
Application
ISTR - Intergalactic Space Travel Request
Active
A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.
Allow Digest
When to send
Who will receive
What it will contain
When using an Email Template, the Subject and Message content is picked from that template unless a different Subject or Message content is specified in this form.
Email template
Subject
ISTR Action Required: Manager Approval - ${number}
Message HTML
Select variables:
Fields
Verdana
8pt
Current State
${state}
Manager
${manager}
Approve Mission
Reject Mission
ISTR Mission Management System
This is an automated notification. Please do not reply directly to this email.
table › tbody › tr › td
Press Alt+0 for help
Preview Notification
Update
Delete

8. Advantages & Disadvantages

Advantages

- Centralizes the entire mission request lifecycle in a single ServiceNow application
- Automates multi-level approvals, reducing manual follow-up and delays
- Enforces strong, granular security through 10 role-based ACLs
- Provides real-time visibility via dashboards and reports
- Automated notifications keep all stakeholders informed without manual effort
- Built on a scalable, configuration-first ServiceNow scoped application, easing future maintenance

Disadvantages / Limitations

- Requires a ServiceNow instance and licensing, which may not be practical outside an enterprise context
- New users face a learning curve with ServiceNow's Service Portal and approval concepts
- Currently web-based only; no dedicated native mobile app
- Dependent on internet connectivity and ServiceNow platform availability
- As a prototype/academic build, it has not been load-tested at real enterprise scale

9. Conclusion

The Intergalactic Space Travel Request (ISTR) System successfully digitizes and automates the end-to-end mission request lifecycle using the ServiceNow platform. By integrating the Service Catalog, Flow Designer, Business Rules, Client Scripts, UI Policies, ACLs, Notifications, Reports, and Dashboards, the application delivers a secure, scalable, and transparent solution for managing intergalactic mission requests. It reduces manual effort, accelerates approvals, improves collaboration among departments, and provides real-time visibility into mission operations, making it a robust enterprise workflow automation solution and a strong demonstration of applied ServiceNow development skills.

10. Future Scope

- AI-based mission risk prediction
- Predictive approval timelines
- Dedicated mobile application
- Voice-assisted request submission
- Integration with external space agencies
- Live mission tracking
- AI chatbot support for requesters
- Advanced analytics and predictive maintenance dashboards
- Multi-language support

11. Appendix

Dataset Link

Not Applicable – ISTR does not use an external dataset; all data is created and managed natively within the ServiceNow custom table during use.

GitHub & Project Demo Link

[GitHub repository link here]

[project demo video/live walkthrough link here]